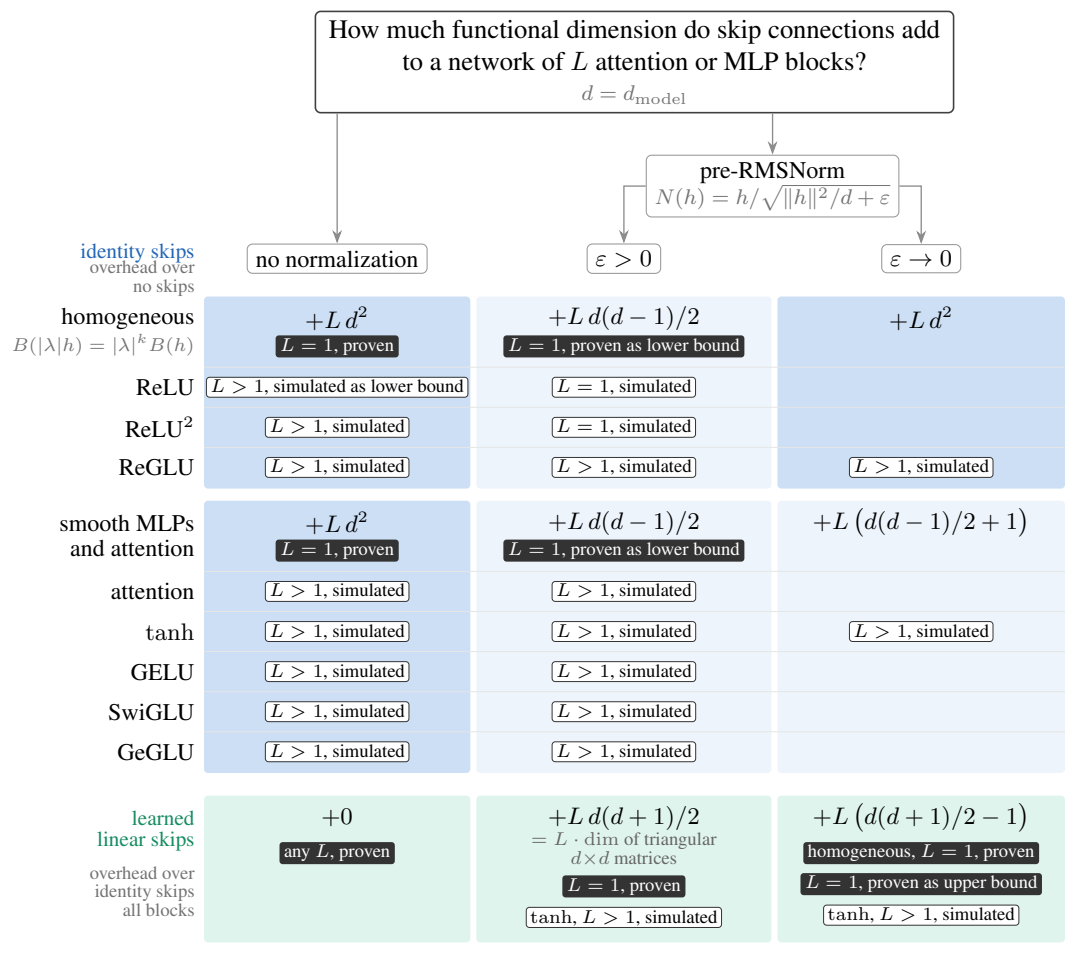

How much functional dimension do skip connections add to a network of L attention or MLP blocks?
$d = d_{\text{model}}$
pre-RMSNorm
$N(h) = h/\sqrt{\|h\|^2/d + \varepsilon}$
identity skips
overhead over no skips
no normalization
$\varepsilon > 0$
$\varepsilon \to 0$
homogeneous
$B(|\lambda|h) = |\lambda|^k B(h)$
$+L\,d^2$
$L = 1$, proven
$+L\,d(d-1)/2$
$L = 1$, proven as lower bound
$+L\,d^2$
ReLU
$L > 1$, simulated as lower bound
$L = 1$, simulated
ReLU^2
$L > 1$, simulated
$L = 1$, simulated
ReGLU
$L > 1$, simulated
$L > 1$, simulated
$L > 1$, simulated
smooth MLPs and attention
$+L\,d^2$
$L = 1$, proven
$+L\,d(d-1)/2$
$L = 1$, proven as lower bound
$+L\left(d(d-1)/2+1\right)$
attention
$L > 1$, simulated
$L > 1$, simulated
tanh
$L > 1$, simulated
$L > 1$, simulated
$L > 1$, simulated
GELU
$L > 1$, simulated
$L > 1$, simulated
SwiGLU
$L > 1$, simulated
$L > 1$, simulated
GeGLU
$L > 1$, simulated
$L > 1$, simulated
learned linear skips
overhead over identity skips
all blocks
$+0$
any L, proven
$+L\,d(d+1)/2$
$= L \cdot$ dim of triangular $d \times d$ matrices
$L = 1$, proven
tanh, $L > 1$, simulated
$+L\left(d(d+1)/2-1\right)$
homogeneous, $L = 1$, proven
$L = 1$, proven as upper bound
tanh, $L > 1$, simulated